UNIVERSITY OF SOUTH DAKOTA

Generative Image Denoising Using Conditional Diffusion Model

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Outline

- Introduction
- Team Contribution
- Model Design and Methodology
- Data Description
- Experiments and Results
- Conclusion
- Future Work
- References

Inspiration and Objective

Inspiration

- Image noise reduces visual quality and affects in medical imaging, photography, and Artificial Intelligence applications.
- Denoising methods like CNNs or autoencoders struggle to retain fine structural details.
- Diffusion models have shown remarkable generative and restoration capabilities, outperforming traditional methods.

Objective

- Develop a conditional diffusion model for image denoising.
- Preserve texture, structure, and details by removing the noise.
- Achieve higher PSNR, SSIM and perceptual quality.

Team Contribution

Name	Section & Level (Graduate)	Contribution
Santosh Luitel	CSC 557 U20	❑ Worked on model training, evaluation, and tuning of diffusion parameters. ❑ Analyzed experimental results and prepared performance comparisons (PSNR, SSIM, LPIPS).
Gaurav Regmi	CSC 557 U20	❑ Handled data preprocessing and augmentation. ❑ Assisted in model implementation using U-Net architecture.
Aavash Silwal	CSC 557 U20	❑ Worked on preparing slides and designing the overall presentation layout. ❑ Helped design the methodology section and worked on coding the diffusion and reverse process and training the model.
Sambhav Siwakoti	CSC 557 U20	❑ Focused and research on diffusion models and their applications. ❑ Helped design the methodology section and documented the diffusion and reverse process.

Model Design and Methodology

- Diffusion models are probabilistic generative models that learn to create data by gradually removing noise.
- A diffusion model comprises two major processes:
  - forward process and
  - reverse process.

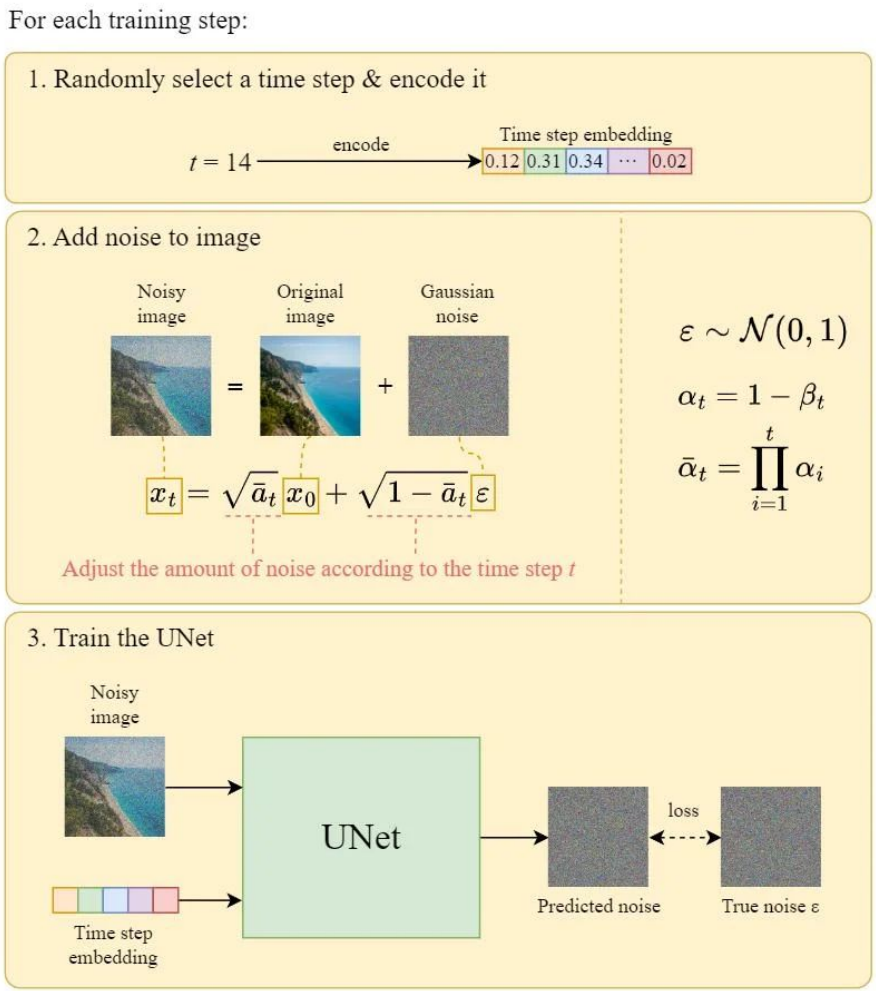
Forward Diffusion Process

- Represented as a Markov chain:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I)$$

- At each step t, small Gaussian noise is added.
- After T steps, the image becomes indistinguishable from pure Gaussian noise.
- The purpose is to define a smooth, tractable degradation path the model can learn to reverse.

Model Design and Methodology



In each epoch:

A random time step t will be selected for each training sample (image).

Apply the Gaussian noise .

Convert the time steps to embeddings (vectors). Time information is important. The model needs to be aware of the specific time step at which it is predicting noise.

https://codoraven.com/blog/ai/diffusion-model-clearly-explained/

Reverse Diffusion Process

- Reverse process aims to achieve the goal opposite to the forward process by denoising to generate a sample.
- The model learns $p_\theta(x_{t-1} | x_t)$ to predict the previous, less-noisy image.
- We start with an initial random noise sample $x_t \sim N(0,I)$ and progressively reduce the noise and gradually restore a sample.
- The purpose is to learn to reverse the noise corruption process.

What the UNet Learns

- The forward diffusion can be done using the closed-form formula.
- The backward diffusion can be done using a trained neural network.(Unet)
- To approximate the desired denoising step q, we just need to approximate the noise ε (t) using a neural network ε(θ).
  - Learns to predict Gaussian noise added during diffusion
  - Inputs:

    - x_t: noisy image

    - y: corresponding real noisy image (conditioning)

    - t: diffusion time step embedding

  - Output: predicted noise $\hat{\varepsilon}(x_t, y, t)$

Dataset

Source: DIV2K Dataset – benchmark for image restoration.

Composition:

- **Training set:** 800 high-resolution images with noisy pairs
- **Test/Validation set:** 100 high-resolution images with noisy pairs

Purpose: Provides diverse natural scenes for training the diffusion model to generalize across textures and lighting.

Data Preprocessing

- **Image size** (256×256 crops)
- **Color images** (3 channels)
- **gaussian_sigmas** = [0.05, 0.1, 0.2]
- **jpeg_qualities** = [90, 70, 50]
- **motion_sizes** = [1, 3, 5]

After filtering, each images are passed into functions like add_gausian_noise(), add_motion_blur, and add_jpeg_artifiact which will generate multiple images with noise.

```
Sample rows from manifest.csv:
split,stem,tag,clean_path,noisy_path,manual_tags
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0.1.png,"{""keep"": true, ""notes"": """}"
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g,"{""keep"": true, ""notes"": """}"
```
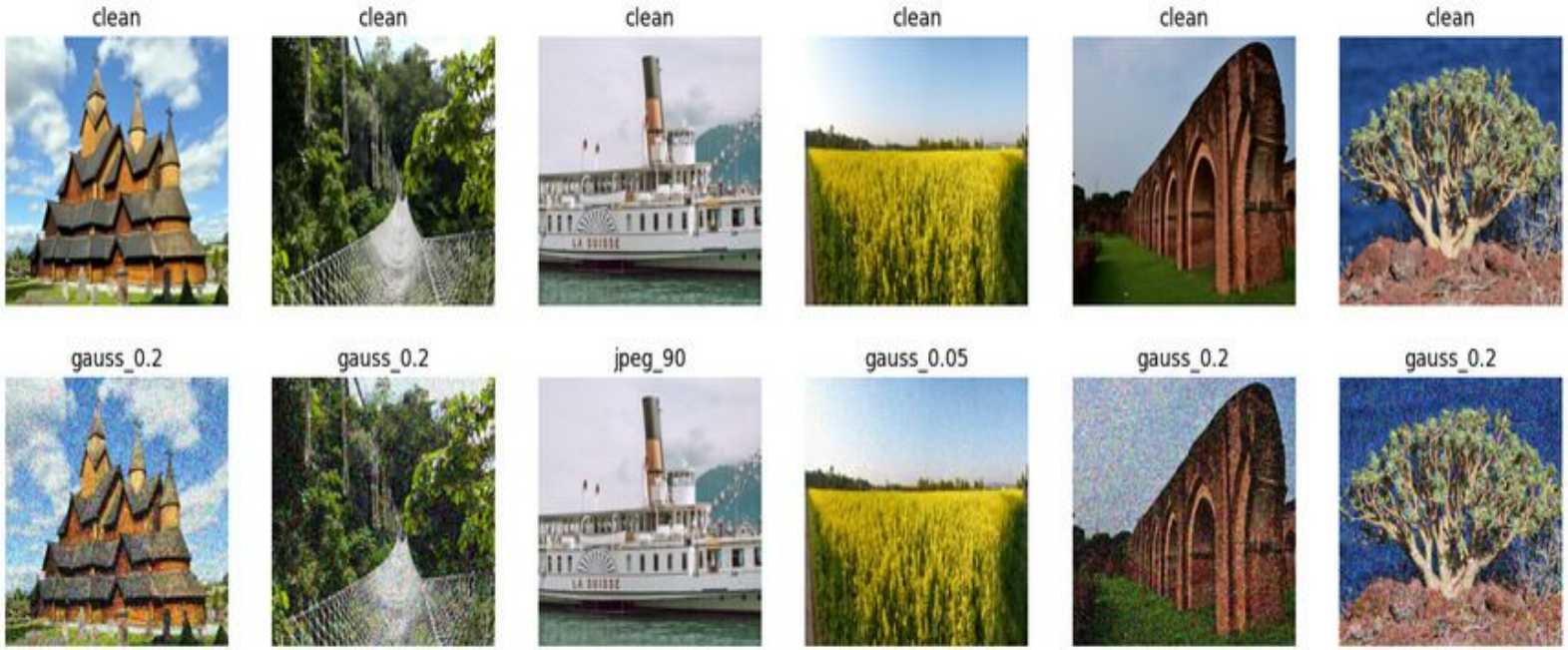
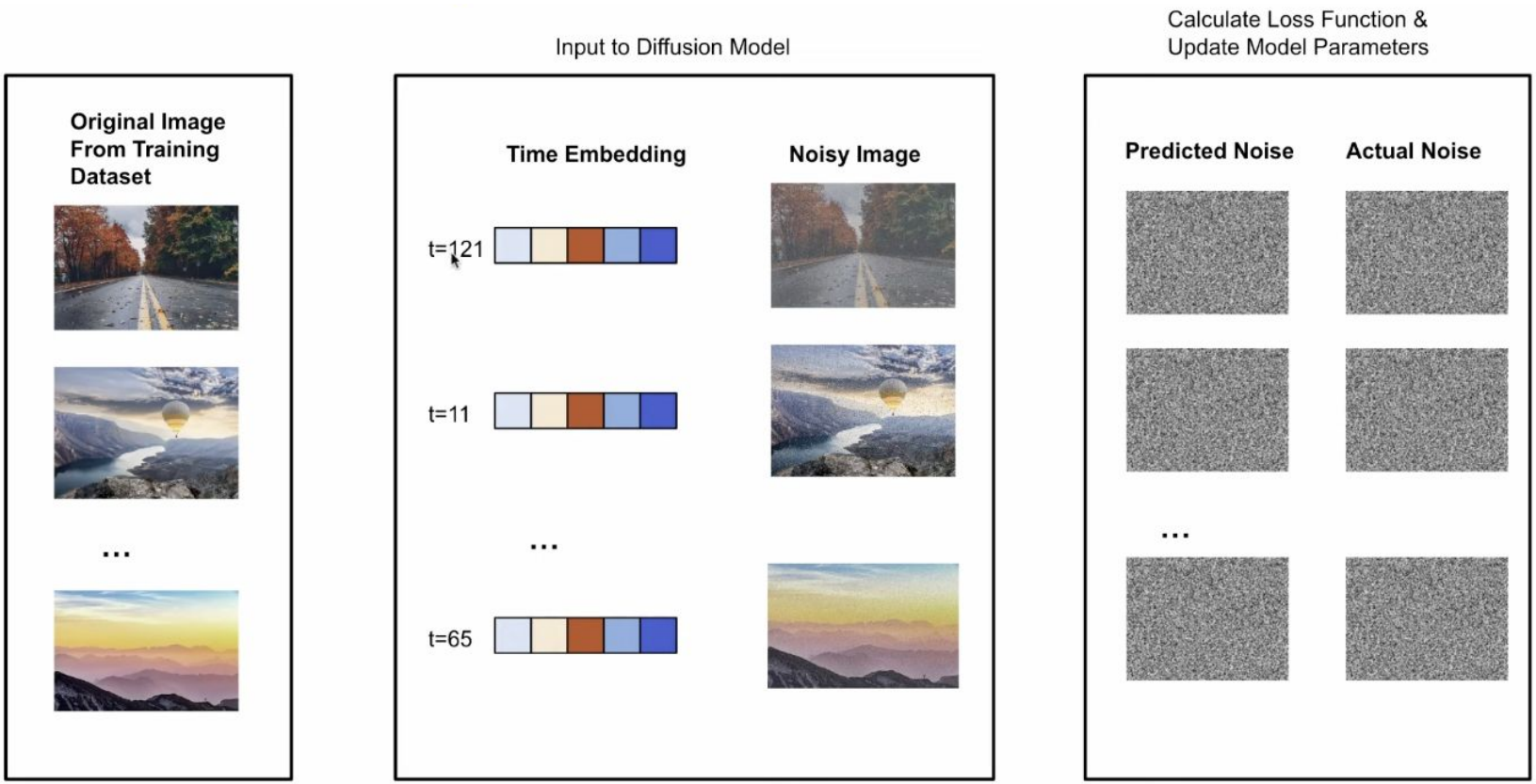

Data Preprocessing



Diffusion Model



Experiments and Results

Goal:

To evaluate a Denoising Diffusion Probabilistic Model (DDPM) on the DIV2K dataset for removing synthetic noise from high-resolution images.

Experimental Setup

- Implemented **DDPM** for image denoising using **DIV2K** dataset.
- **Dataset split:** 7,200 train | 900 validation.
- **Image size:** 256×256 (RGB)
- **Diffusion steps:** 1000
- **Noise types:** Gaussian, JPEG compression, and Motion blur.
- **Training:** 100 epochs | LR = 2e-4 | Loss = MSE | Optimizer = Adam.
- **Environment:** Python, PyTorch, LPIPS, Google Colab (GPU).

Evaluation Metrics

- **PSNR (Peak Signal-to-Noise Ratio)**
  Measures pixel-level similarity between denoised and original images.
  Higher PSNR - better reconstruction quality.

  20 dB < N ≤ 30 dB = Good, **> 30 dB = Excellent**

- **SSIM (Structural Similarity Index)**
  Evaluates image quality based on structure, contrast, and luminance.

  Value closer to 1 higher structural similarity.

- **LPIPS (Learned Perceptual Image Patch Similarity)**
  Uses deep features to measure perceptual similarity.

  Lower LPIPS - more visually realistic output.

Evaluation Metrics

Epoch Table:

Epoch	Train Loss	PSNR (dB)	SSIM	LPIPS
12	0.0338	10.99	0.158	0.673
14	0.0330	17.86	0.606	0.4001
85	0.0307	24.47	0.779	0.173
100	0.0279	16.28	0.468	0.456

Inference Results

Model: Best checkpoint (Epoch 85)
Validation Dataset: 900 noisy images

Metric	Inference Value	Best Training (Epoch 85)	Gap
PSNR	**21.25 dB**	24.47 dB	-3.22 dB
SSIM	**0.611**	0.779	-0.168
LPIPS	**0.3293**	0.1730	+0.1563

Inference Results

Interpretation:

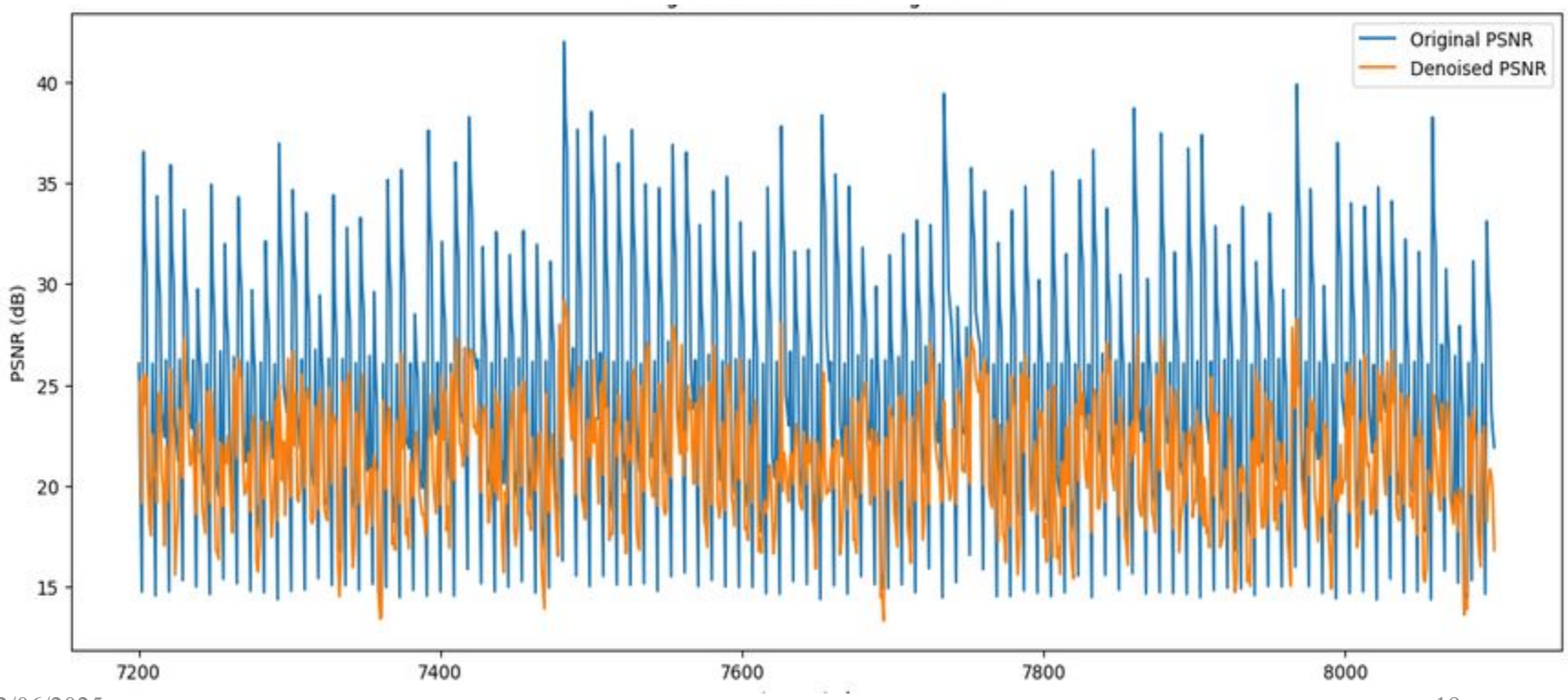
- Strong inference performance on validation set
- Model generalizes well
- All 900 images successfully denoised
- Results indicate robust model for real-world denoising tasks

Evaluation Metrics

Evaluation Metrics on Validation Set (900 Images)

PSNR

Definition: Measures signal power relative to noise power



Evaluation Metrics

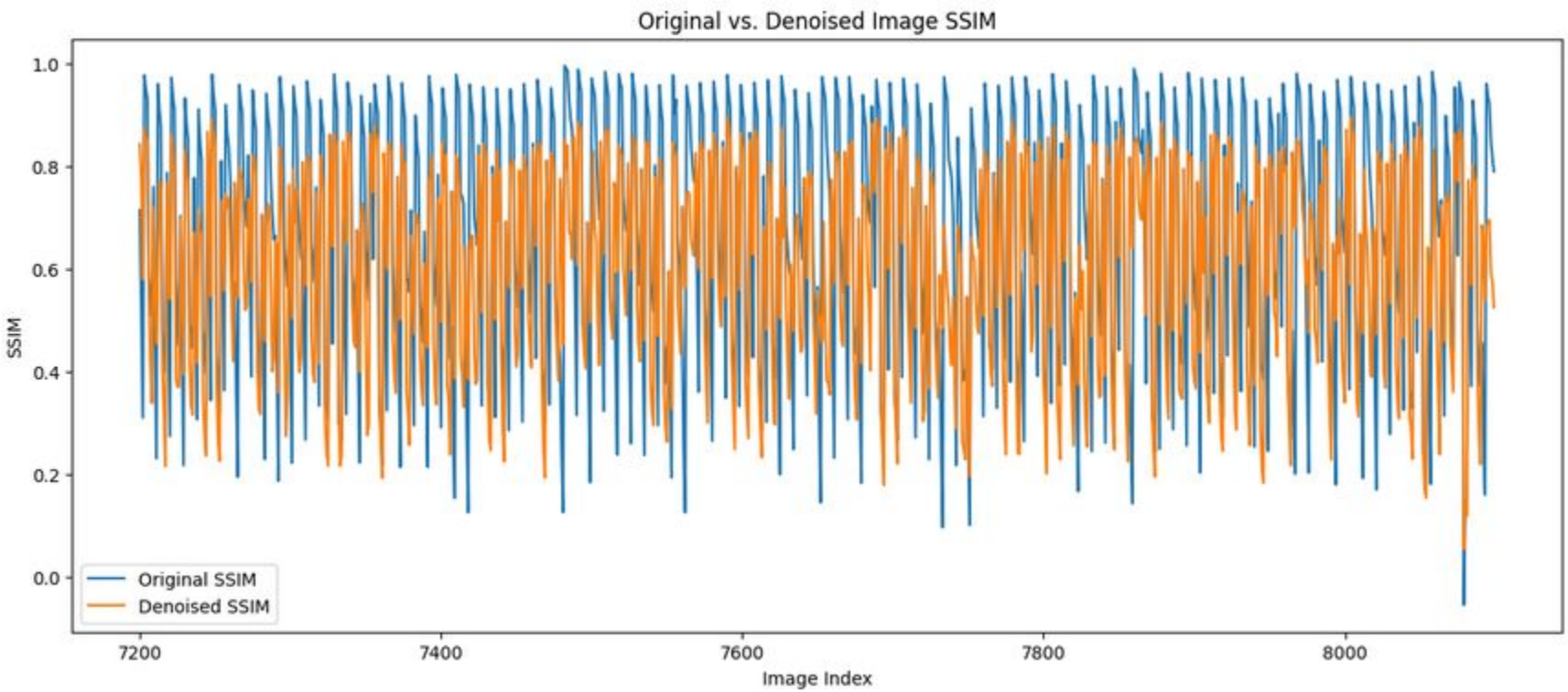
From the PSNR Chart:

- **Blue line** (Original): ~24.18 dB - noisy reference baseline
- **Orange line** (Denoised): 21.25 dB - against clean ground truth
- **Variance:** Noisy images show high variance; denoised images more stable
- **Result:** 21.25 dB is competitive quality for denoising tasks

Evaluation Metrics

SSIM (Structural Similarity Index): 0.611 (Scale: 0-1, where 1 = identical)

Definition: Measures structural similarity including luminance, contrast, and structure



Evaluation Metrics

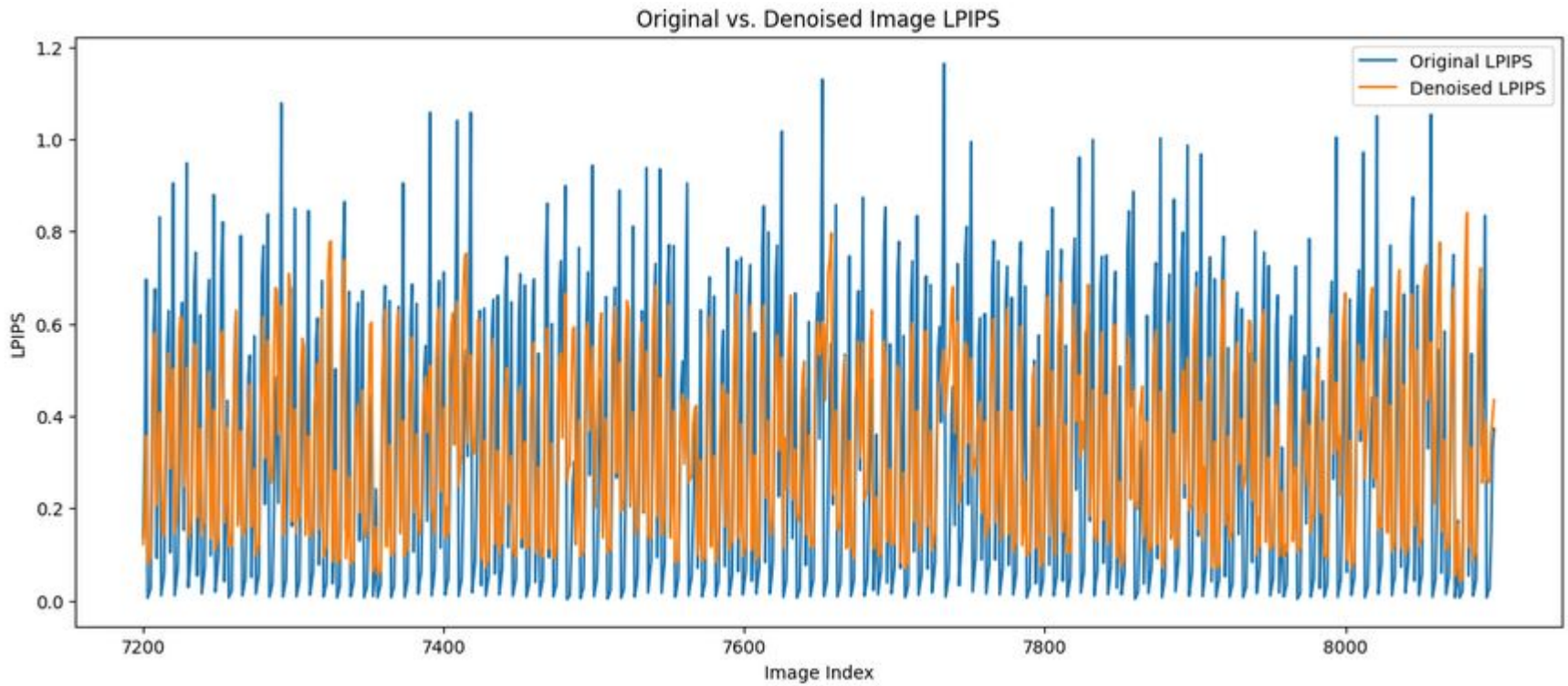
From the SSIM Chart:

- **Blue line** (Original): High variance (0.1-1.0) - inconsistent structure.
- **Orange line** (Denoised): Stable range (0.6-0.8) - consistent structure.
- **Key Finding:** Denoised outputs preserve structure while removing noise.
- **0.611 value:** Good structural alignment with ground truth.

Evaluation Metrics

LPIPS (Learned Perceptual Image Patch Similarity): 0.3293 (Scale: 0-1, where 0 = identical)

Definition: AI-based perceptual metric that matches human visual perception



Evaluation Metrics

From the LPIPS Chart:

- **Blue line** (Original): High variance (0.0-1.2) - far from clean images
- **Orange line** (Denoised): Lower range (0.2-0.8) - closer to ground truth
- **Key Finding:** Denoised images show LOWER perceptual distance
- **0.3293 value:** Images are perceptually similar to clean reference

Conclusion

- Developed a Conditional Diffusion Model for effective image denoising.
- Successfully removed various types of noise, such as Gaussian, motion blur, and JPEG artifacts.
- Model preserved texture and structural details better than traditional CNN-based denoisers.
- Achieved best results at Epoch 85 with PSNR = 24.47 dB, SSIM = 0.779, and LPIPS = 0.1730.

Future Works

- Extend testing to real-world noisy datasets beyond synthetic noise.
- Tune the β-schedule and apply cosine/noise variance optimization for smoother diffusion.
- Integrate perceptual or adversarial losses to enhance visual quality.
- Compare against state-of-the-art denoisers for comprehensive performance benchmarking.

References

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- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). “High-resolution image synthesis with latent diffusion models.”

Thank You!

Any queries, comments, or feedback?